Folded Transport MCMC: Certified Sampling of Symmetric Multi-Modal Posteriors via Quotient-Space Reduction

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Abstract

Transport maps trained as normalising flows provide powerful independence proposals for Markov chain Monte Carlo, and recent work has established certified spectral-gap lower bounds for such chains. However, when the target posterior possesses a finite symmetry group—label switching in mixture models, reflection symmetry, or permutation invariance in structural identification—the flow must represent multiple separated modes simultaneously, inflating the oscillation bound and rendering the certificate vacuous. We introduce *Folded Transport MCMC* (FolT-MCMC), which eliminates symmetric multimodality by constructing a quotient-space independence sampler on the fundamental domain of the symmetry group. We prove that the LCNF certification framework transfers to the quotient metric with improved covering radius and density floor, and that the certified spectral-gap lower bound improves whenever the unfolded flow exhibits cross-mode proposal deficiency. On Gaussian mixtures (d = 2–20) and label-switching targets (up to 24 equivalent modes), the certified improvement ratio ranges from 2× to 145×, with the folded certificate empirically nearly dimension-free. On real accelerometer data from a supertall building during Typhoon Mangkhut, FolT-MCMC yields a non-vacuous quantile-core certificate for a simplified closely-spaced three-mode identification problem, where the unfolded certificate is vacuous.

Keywords: Transport MCMC; normalising flows; spectral gap; label switching; symmetry; Bayesian structural identification.

1 Introduction

Transport maps trained as normalising flows have become a powerful tool for accelerating Markov chain Monte Carlo [Parno and Marzouk, 2018, Hoffman et al., 2019, Gabrié et al., 2022]. By learning a diffeomorphism that approximately Gaussianises the target density, a transport map provides a high-quality independence proposal whose mismatch with the target can be corrected exactly through Metropolis–Hastings acceptance. Recent work has established *certified* spectral-gap lower bounds for such chains via computable oscillation bounds on the log-density ratio [Hu, 2026b,a], giving practitioners a provable convergence guarantee rather than a heuristic diagnostic.

A fundamental limitation of existing transport MCMC and its certification framework is *symmetric multimodality*. When the posterior possesses a finite symmetry group—label switching in mixture models (m! equivalent modes), reflection symmetry in latent-variable models, or approximate permutation invariance in structural modal identification—the flow must represent multiple separated modes simultaneously. Finite-capacity architectures such as coupling-based RealNVP struggle with this task, especially in higher dimensions. The resulting cross-mode oscillation in the log-density ratio inflates the certified bound and can render the spectral-gap certificate vacuous, even when the flow fits each individual mode well.

This paper introduces *Folded Transport MCMC* (FolT-MCMC), which eliminates symmetric multimodality before sampling by passing to a quotient target on a fundamental domain of the symmetry group. The method constructs a quotient-space independence sampler whose symmetrised proposal density removes the cross-mode oscillation, while the LCNF covering certificate transfers to the quotient metric with full ball-volume preservation. The three main contributions are:

1. **Quotient-space framework.** We define the folded target $\pi_F = |G| \cdot \pi$ on the fundamental domain, the quotient proposal $q_F(z) = \sum_{g \in G} q_{T_F}(gz)$, and the associated quotient metric d_G. The HPD set satisfies an exact identity $\mathcal{K}_\alpha^F = \mathcal{K}_\alpha \cap D$, and the density floor improves by a factor of $|G|$ (Section 3).
2. **Certified improvement.** We prove that the LCNF empirical oscillation theorem carries over to the quotient setting, that the covering radius does not increase ($\varepsilon_F^* \leq \varepsilon_U^*$), and that the certified spectral-gap lower bound improves whenever the unfolded flow exhibits cross-mode proposal deficiency—a condition satisfied empirically in all our experiments (Section 4).
3. **Empirical validation.** On Gaussian mixtures ($d = 2$–20) and label-switching targets ($m! =$ 2–24 modes), the quantile-core certified lower bound ratio $\underline{\gamma}_F/\underline{\gamma}_U$ ranges from $2\times$ to $145\times$. The folded certificate is empirically nearly dimension-free, while the unfolded certificate collapses with both dimension and mode count. On real accelerometer data from a supertall building during Typhoon Mangkhut, FolT-MCMC yields a non-vacuous quantile-core certificate for a simplified closely-spaced three-mode identification problem, where the unfolded certificate is vacuous (Sections 5–6).

The remainder of the paper is organised as follows. Section 2 reviews transport MCMC, the LCNF certification framework, and symmetry in Bayesian inference. Section 3 develops the FolT-MCMC framework. Section 4 establishes the theoretical guarantees. Section 5 presents the synthetic experiments. Section 6 applies FolT-MCMC to structural modal identification with real typhoon data. Section 7 discusses limitations and extensions.

2 Background

2.1 Transport MCMC and independence Metropolis–Hastings

A normalising flow $T\colon \mathbb{R}^d \to \mathbb{R}^d$ is a learnable diffeomorphism that pushes a simple base density q_0 (typically $\mathcal{N}(0, I_d)$) to an approximation $q = T_\# q_0$ of a target density π. In transport MCMC, the flow serves as the proposal mechanism for a Markov chain whose stationary distribution is exactly π, correcting any approximation error in q through the Metropolis–Hastings acceptance step [Parno and Marzouk, 2018, Hoffman et al., 2019, Gabrié et al., 2022].

The simplest such chain is *independence Metropolis–Hastings* (IMH): given the current state θ, propose $\theta' \sim q$ and accept with probability $\min(1, e^{h(\theta)-h(\theta')})$, where $h = \log \pi - \log q$ is the log-density ratio. IMH is particularly well suited to transport proposals because the proposal is cheap to evaluate (a single forward pass through the flow) and the acceptance probability depends only on the quality of the density-ratio fit.

The spectral gap γ of the IMH chain—measuring the rate of geometric convergence—is bounded below by the Mengersen–Tweedie inequality [Mengersen and Tweedie, 1996]:

$$\gamma \;\geq\; \frac{2}{1+\exp\bigl(\operatorname{osc}(h)\bigr)}, \qquad \operatorname{osc}(h) := \sup_\theta h(\theta) - \inf_\theta h(\theta). \tag{1}$$

When the flow approximation is perfect ($q = \pi$), $\mathrm{osc}(h) = 0$ and $\gamma = 1$; as the approximation degrades, $\mathrm{osc}(h)$ grows and γ shrinks exponentially.

2.2 Certified spectral gaps via LCNF

The oscillation $\mathrm{osc}(h)$ is not directly computable from finite samples, because both the supremum and infimum range over the entire support of π. The LCNF framework [Hu, 2026b] provides a *certified* upper bound on $\mathrm{osc}(h)$, and thereby a certified lower bound on γ, using three ingredients.

Spectrally normalised RealNVP. The flow T is built from L coupling layers, each with a spectral-norm constraint on the conditioner network and a scale clip c. This gives a per-layer bi-Lipschitz bound, ensuring that the Jacobian and its log-determinant are well controlled.

Empirical oscillation theorem. Given n i.i.d. samples $\{Z_i\}$ from π and the $(1-\alpha)$-HPD set $\mathcal{K}_\alpha$, the LCNF theorem states that with probability $\geq 1 - \delta$:

$$\mathrm{osc}_{\mathcal{K}_\alpha}(h) \;\leq\; \underbrace{\max_i h(Z_i) - \min_i h(Z_i)}_{\widehat{\mathrm{osc}}_n} \;+\; \underbrace{2\,M\,\varepsilon^*}_{\text{interpolation error}}, \tag{2}$$

where $M = \sup_{\mathcal{K}_\alpha} \|\nabla h\|$ is the local gradient constant and ε^* is the minimal covering radius satisfying a covering condition that depends on $\mathrm{diam}(\mathcal{K}_\alpha)$, $\pi_{\min}$, and n. The right-hand side is fully computable from samples and network parameters.

Quantile-core certificates. The full oscillation bound (2) is often dominated by extreme tails. The CerT-MCMC extension [Hu, 2026a] replaces the full HPD set with a ρ-trimmed core that excludes the highest-oscillation fraction, yielding tighter *quantile-core* certificates. In all experiments of the present paper, we report quantile-core certificates at $\rho = 0.05$.

Limitation: multimodality. When π has k well-separated modes and the flow can only Gaussianise one of them, the log-density ratio h is large on the off-modes (where $q \ll \pi$), inflating $\widehat{\mathrm{osc}}_n$ by a cross-mode gap that grows with mode separation and dimension. This is the fundamental barrier that FolT-MCMC is designed to overcome.

2.3 Symmetry in Bayesian inference

Many Bayesian posteriors possess a finite symmetry group G under which the likelihood and prior are jointly invariant. The resulting symmetric multimodality is a well-known obstacle to MCMC mixing and inference.

Label switching. In m-component mixture models, the likelihood is invariant under permutation of component labels ($G = S_m$, $|G| = m!$). This creates $m!$ equivalent posterior modes, between which standard MCMC chains mix poorly [Celeux et al., 2000, Jasra et al., 2005]. Existing solutions are predominantly *post-hoc*: after running MCMC in the full $m!$-modal space, the samples are relabelled using a loss function [Stephens, 2000], an equivalence-class representative algorithm [Papastamoulis and Iliopoulos, 2010], or random permutation at each step [Frühwirth-Schnatter, 2001]. These methods do not address the poor mixing of the underlying chain, nor do they provide certified convergence guarantees.

Structural modal identification. In Bayesian operational modal analysis of buildings and bridges, closely-spaced structural modes create approximate label-switching symmetry: the posterior over m sets of modal parameters (frequency, damping ratio) is nearly S_m-invariant when the modes have similar frequencies [Au, 2011, Lam et al., 2019]. This is a key motivation for the present work.

Equivariant normalising flows. An alternative approach constructs flows that respect the symmetry by design: $T(g \cdot z) = g \cdot T(z)$ [Köhler et al., 2020, Rezende et al., 2019]. Such equivariant flows reduce parameter redundancy but must still represent all $|G|$ modes. FolT-MCMC takes the complementary approach of *eliminating* the symmetric modes before sampling, reducing the target to a single representative per orbit. The two strategies are orthogonal and could in principle be combined.

3 The FolT-MCMC framework

Let π be a target density on $\mathbb{R}^d$ that is invariant under a finite group G acting on $\mathbb{R}^d$: $\pi(g \cdot \theta) = \pi(\theta)$ for every $g \in G$ and almost every θ. Write $s = |G|$. Typical examples are label-switching symmetry in mixture models ($G = S_m$, the symmetric group on m components, $s = m!$) and reflection symmetry ($G = \mathbb{Z}_2$, $s = 2$).

The central idea of FolT-MCMC is to *eliminate* the symmetric multimodality of π before sampling, by passing to a quotient target on a fundamental domain, running transport-preconditioned independence Metropolis–Hastings there, and certifying the resulting chain with the LCNF oscillation framework.

3.1 Quotient target

A *fundamental domain* for G is a measurable set $D \subset \mathbb{R}^d$ such that $\mathbb{R}^d = \bigcup_{g \in G} g \cdot D$ with pairwise disjoint interiors and $\pi(g \cdot D \cap g' \cdot D) = 0$ for $g \neq g'$.

Definition 3.1 (Quotient target)**.** The *folded* or *quotient target* on D is

$$\pi_F(z) \;=\; s \cdot \pi(z), \qquad z \in D. \tag{3}$$

The corresponding potential is $U_F(z) = -\log \pi_F(z) = U(z) - \log s$.

Normalisation is immediate: by the fundamental-domain decomposition, $\int_D \pi_F = s \int_D \pi = s \cdot (1/s) = 1$. The density π_F retains one representative from each G-orbit of modes, so the symmetric multimodality induced by the group action is absent; non-symmetric multimodality, if present, may remain.

The next result shows that folding preserves the highest posterior density (HPD) structure exactly.

Proposition 3.2 (HPD identity)**.** *Let u_α be the $(1-\alpha)$-quantile of $U(\Theta)$ under $\Theta \sim \pi$, defining the HPD set $\mathcal{K}_\alpha = \{\theta \in \mathbb{R}^d : U(\theta) \leq u_\alpha\}$. Then:*

1. *$u_\alpha^F = u_\alpha - \log s$;*
2. *$\mathcal{K}_\alpha^F := \{z \in D : U_F(z) \leq u_\alpha^F\} = \mathcal{K}_\alpha \cap D$ (a.e.).*

Proof. For any Borel function $\varphi \colon \mathbb{R} \to \mathbb{R}$, the G-invariance of π and the fundamental-domain decomposition give

$$\int_{\mathbb{R}^d} \varphi(U(\theta))\, \pi(\theta)\, \mathrm{d}\theta = \sum_{g \in G} \int_{gD} \varphi(U(\theta))\, \pi(\theta)\, \mathrm{d}\theta = s \int_D \varphi(U(z))\, \pi(z)\, \mathrm{d}z = \int_D \varphi(U(z))\, \pi_F(z)\, \mathrm{d}z.$$

Hence $U(Z_F)$ under $Z_F \sim \pi_F$ has the same distribution as $U(\Theta)$ under $\Theta \sim \pi$, so the two quantiles coincide: the $(1-\alpha)$-quantile of $U_F(Z_F) = U(Z_F) - \log s$ is $u_\alpha - \log s$. Part (ii) follows directly: $\{U_F \le u_\alpha^F\} = \{U - \log s \le u_\alpha - \log s\} = \{U \le u_\alpha\}$. $\square$

Corollary 3.3. *(a)* $\operatorname{diam}(\mathcal{K}_\alpha^F) \le \operatorname{diam}(\mathcal{K}_\alpha)$. *(b) The density floor satisfies* $\pi_{\min}^F := \inf_{\mathcal{K}_\alpha^F} \pi_F = s \cdot \inf_{\mathcal{K}_\alpha \cap D} \pi \ge s \cdot \pi_{\min}$, *where* $\pi_{\min} := \inf_{\mathcal{K}_\alpha} \pi$.

Using essential infima with respect to the Lebesgue measure, the G-invariance of π ensures $\operatorname{ess\,inf}_{\mathcal{K}_\alpha \cap D} \pi = \operatorname{ess\,inf}_{\mathcal{K}_\alpha} \pi$, so $\pi_{\min}^F = s\,\pi_{\min}$. With ordinary infima this equality holds for closed fundamental domains; in general we use the conservative bound $\pi_{\min}^F \ge s\,\pi_{\min}$. For label switching with $m = 4$ components, $s = 24$ and the floor is 24 times higher.

3.2 Quotient proposal

Let $T_F \colon \mathbb{R}^d \to \mathbb{R}^d$ be a normalising flow (e.g. spectrally normalised RealNVP) trained on folded samples projected onto D, and let $q_{T_F} = (T_F)_\# \mathcal{N}(0, I_d)$ denote the induced density on $\mathbb{R}^d$.

Definition 3.4 (Quotient proposal)**.** The *quotient proposal* on D is

$$q_F(z) \;=\; \sum_{g \in G} q_{T_F}(g \cdot z), \qquad z \in D. \tag{4}$$

Normalisation follows from the fundamental-domain decomposition: $\int_D q_F = \sum_g \int_D q_{T_F}(g \cdot z)\,\mathrm{d}z = \sum_g \int_{gD} q_{T_F}(\theta)\,\mathrm{d}\theta = \int_{\mathbb{R}^d} q_{T_F} = 1$. The construction is analogous to Schur averaging in representation theory: q_F is the unique G-symmetric density on $\mathbb{R}^d$ that agrees with the orbit sum of q_{T_F} when restricted to D.

The log-density ratio governing the Metropolis–Hastings acceptance is

$$h_F(z) \;=\; \log \frac{\pi_F(z)}{q_F(z)} \;=\; \log\bigl(s \cdot \pi(z)\bigr) \;-\; \log\Bigl(\sum_{g \in G} q_{T_F}(g \cdot z)\Bigr). \tag{5}$$

Remark 3.5 (Relation to unfolded MH). Independence MH on D with target π_F and proposal q_F is a valid Markov chain with stationary distribution π_F. It is *not* identical to running MH on $\mathbb{R}^d$ and projecting samples onto D, because $\pi_F(z)/q_F(z)$ generally differs from $\pi(z)/q_{T_F}(z)$. The two coincide when q_{T_F} is itself G-invariant. In practice, when q_{T_F} concentrates on a single mode, the off-mode terms $q_{T_F}(g \cdot z)$ for $g \ne e$ are negligible for z deep in D, and the two ratios agree approximately.

3.3 Algorithm

FolT-MCMC combines folding, transport, and certification in a three-stage pipeline.

Algorithm: FolT-MCMC

Input: target π with known symmetry group G, fundamental domain D.

Stage 1 (Fold and train).

Draw reference samples $\{z_i\}_{i=1}^N$ from π_F (fold exact or MCMC samples into D).

Train a spectrally normalised RealNVP flow T_F on π_F using NLL + oscillation regularisation.

Form the quotient proposal q_F via (4).

Stage 2 (Sample).

Run independence Metropolis–Hastings on D:

Propose $z' \sim q_F$ by sampling $z_0 \sim \mathcal{N}(0, I)$, computing $\theta = T_F(z_0)$, setting $z' = \mathrm{proj}_D(\theta)$.

Accept with probability $\min\big(1,\, e^{h_F(z_{\mathrm{curr}}) - h_F(z')}\big)$.

Stage 3 (Certify).

Compute the empirical oscillation $\widehat{\mathrm{osc}}_n^F$ and gradient supremum $\widehat{M}_n^F$ from an independent certification set on D.

Solve the covering condition (9) for ε_F^*.

Report the certified spectral-gap lower bound $\underline{\gamma}_F \geq 2/(1 + \exp(\widehat{\mathrm{osc}}_n^F + 2\widehat{M}_n^F \varepsilon_F^*))$.

Output: certified posterior samples on D; spectral-gap certificate $\underline{\gamma}_F$.

We highlight two concrete instantiations.

Reflection fold ($G = \mathbb{Z}_2$). The non-identity element acts by negating the first coordinate: $g \cdot \theta = (-\theta_1, \theta_2, \ldots, \theta_d)$. The fundamental domain is $D = \{\theta_1 \geq 0\}$ and $\mathrm{proj}_D(\theta) = (|\theta_1|, \theta_2, \ldots, \theta_d)$. The quotient proposal reduces to $q_F(z) = q_{T_F}(z) + q_{T_F}(-z_1, z_2, \ldots, z_d)$.

Permutation fold ($G = S_m$). Each $\theta \in \mathbb{R}^d$ is partitioned into m blocks of p parameters ($d = mp$), and $G = S_m$ permutes blocks. The fundamental domain is the sorted chamber $D = \{\theta : \theta_1^{(1)} \leq \theta_1^{(2)} \leq \cdots \leq \theta_1^{(m)}\}$, where $\theta_1^{(j)}$ is the first parameter (e.g. natural frequency) of the jth block. Projection sorts the blocks by this parameter.

3.4 Quotient metric and covering

The certification framework of Hu [2026b] relies on a covering lemma (Lemma 5.2 therein) that lower-bounds the probability mass of balls intersected with the HPD set. On the fundamental domain D, Euclidean balls near the fold boundary ∂D may be truncated, potentially losing mass. We resolve this by working in the *quotient metric*.

Definition 3.6 (Quotient metric)**.** For $\theta, \theta' \in \mathbb{R}^d$, define

$$d_G(\theta, \theta') \;=\; \min_{g \in G} \|\theta - g \cdot \theta'\|. \tag{6}$$

This descends to a well-defined metric on the orbit space $\mathbb{R}^d/G$.

Since $d_G \leq \|\cdot\|$ (take $g = e$), any Euclidean ε-cover is automatically a quotient ε-cover, giving the covering-number bound

$$\mathcal{N}_G(\mathcal{K}_\alpha^F, \varepsilon) \;\leq\; \mathcal{N}_E(\mathcal{K}_\alpha^F, \varepsilon) \;\leq\; \Big(\frac{2\mathrm{diam}(\mathcal{K}_\alpha^F)}{\varepsilon} + 1\Big)^d. \tag{7}$$

The advantage of the quotient metric emerges when bounding the *ball mass*. For a point $z \in D$ whose stabiliser is trivial ($\mathrm{Stab}(z) = \{e\}$), the quotient ball $B_G(z, \varepsilon) := \{z' : d_G(z, z') \le \varepsilon\}$ corresponds in $\mathbb{R}^d$ to the union $\bigcup_{g \in G} \big(B(g \cdot z, \varepsilon) \cap g \cdot D\big)$, which has total volume equal to that of a full Euclidean ball $V_d\, \varepsilon^d$—no half-ball or boundary correction is needed. For well-separated mixture posteriors, the HPD set $\mathcal{K}_\alpha^F$ lies in the interior of D and all stabilisers are trivial; we formalise this as a regularity condition.

Assumption 3.7 (Generic stabiliser)**.** $|\mathrm{Stab}(z)| = 1$ *for every* $z \in \mathcal{K}_\alpha^F$*.*

Assumption 3.7 holds whenever the mode centres are distinct after projection into D and the modes are sufficiently well separated that the HPD set does not touch the fold boundary. When it fails (e.g. at fixed-point strata of the group action), the ball-mass bound acquires a correction factor $1/h_{\max}$ where $h_{\max} = \max_{z \in \mathcal{K}_\alpha^F} |\mathrm{Stab}(z)|$; see Supplement A.

The remaining ingredient is a Lipschitz property of h_F under the quotient metric.

Lemma 3.8 (Quotient Lipschitz constant)**.** *Let h_F be extended to $\mathbb{R}^d$ by $h_F(\theta) = \log\big(s\pi(\theta)\big) - \log\big(\sum_g q_{T_F}(g\theta)\big)$, and let $M_F = \sup\big\{\|\nabla h_F(w)\| : d_G(w, \mathcal{K}_\alpha^F) \le \varepsilon_F^*\big\}$. Then for all $x, y \in \mathcal{K}_\alpha^F$:*

$$|h_F(x) - h_F(y)| \;\le\; M_F \cdot d_G(x, y). \tag{8}$$

Proof. Both π and $\sum_g q_{T_F}(g\,\cdot)$ are G-invariant by construction, so $h_F(g \cdot \theta) = h_F(\theta)$ for all g. For any $g \in G$, the mean value theorem gives $|h_F(x) - h_F(y)| = |h_F(x) - h_F(g \cdot y)| \le M_F\, \|x - g \cdot y\|$, where the supremum is taken over the ε_F^*-neighbourhood of $\mathcal{K}_\alpha^F$ to cover the interpolation path. Minimising over g yields (8). □

With the covering number (7), the ball-mass bound, and Lemma 3.8, the LCNF empirical oscillation theorem [Hu, 2026b, Theorem 5.1] carries over to the quotient setting. The covering condition becomes

$$\left(\frac{D_F}{\varepsilon} + 1\right)^d \cdot \big(1 - (s/h_{\max})\, \pi_{\min}^D\, V_d\, \varepsilon^d\big)^{n_{\mathrm{eff}}} \;\le\; \frac{\delta}{3}, \tag{9}$$

where $D_F = \mathrm{diam}(\mathcal{K}_\alpha^F)$, $\pi_{\min}^D = \inf_{\mathcal{K}_\alpha \cap D} \pi$, and $h_{\max} = \max_{z \in \mathcal{K}_\alpha^F} |\mathrm{Stab}(z)|$. Under Assumption 3.7, $h_{\max} = 1$ and the mass factor simplifies to $\pi_{\min}^F \cdot V_d \cdot \varepsilon^d$. The resulting spectral-gap certificate is developed in Section 4.

Remark 3.9 (Computational scalability)*.* The orbit sum in (4) has $s = |G|$ terms. For S_m with $m \le 5$ ($s \le 120$), this is inexpensive. For larger m, the sum can be approximated by restricting to permutations that lie within a neighbourhood of the identity, since distant permutations contribute negligibly when the modes are well separated.

4 Theoretical guarantees

We now establish that the LCNF certification framework [Hu, 2026b] transfers to the quotient setting of Section 3, and that the resulting certified spectral-gap lower bound is at least as sharp as—and typically much sharper than—the bound obtained on the original space.

4.1 Folded LCNF certificate

The LCNF empirical oscillation theorem [Hu, 2026b, Theorem 5.1] requires three regularity conditions on the target, the HPD set, and the log-density ratio. We verify each in turn for the quotient setting $(\pi_F, \mathcal{K}_\alpha^F, h_F)$.

LCNF Assumption 1 (Compact HPD set with positive density floor). By Proposition 3.2, $\mathcal{K}_\alpha^F = \mathcal{K}_\alpha \cap D$ is compact (intersection of a compact set and a closed domain). Its diameter satisfies $D_F \le D_U := \operatorname{diam}(\mathcal{K}_\alpha)$ (Corollary 3.3a), and its density floor satisfies $\pi_{\min}^F \ge s\,\pi_{\min} > 0$ (Corollary 3.3b).

LCNF Assumption 2 (Smooth HPD boundary). The boundary $\partial\mathcal{K}_\alpha^F$ consists of three parts: (i) the HPD level set $\{U_F = u_\alpha^F\} \cap \operatorname{int}(D)$, which is a smooth $(d-1)$-dimensional manifold wherever $\nabla U_F \ne 0$ (holding generically by Sard's theorem); (ii) the fold boundary $\partial D \cap \operatorname{int}(\mathcal{K}_\alpha^F)$, which inherits the smoothness of ∂D (a hyperplane for reflections, a polyhedral cone for permutations); and (iii) the corner set $\{U_F = u_\alpha^F\} \cap \partial D$, which has Hausdorff dimension at most $d-2$ and does not affect the covering argument. Under Assumption 3.7, parts (ii) and (iii) are empty and $\partial\mathcal{K}_\alpha^F$ is smooth.

LCNF Assumption 3 (Local regularity of h_F). On $\operatorname{int}(D)$, $\pi_F = s\pi$ inherits the smoothness of π (typically C^∞ for Gaussian mixtures), and $q_F = \sum_g q_{T_F}(g\,\cdot)$ is C^∞ when T_F uses tanh activations. Therefore $h_F = \log\pi_F - \log q_F$ is C^1 on a neighbourhood of $\mathcal{K}_\alpha^F$, and the gradient supremum M_F (Definition in Lemma 3.8) is finite.

With all three conditions verified, we state the folded certificate.

Theorem 4.1 (FolT-MCMC certificate)**.** *Suppose π is G-invariant with $|G| = s$, Assumption 3.7 holds, and h_F is C^1 on a neighbourhood of $\mathcal{K}_\alpha^F$. Draw $Z_1, \dots, Z_n \overset{\text{iid}}{\sim} \pi_F$ and let $\widehat{\operatorname{osc}}_n^F = \max_i h_F(Z_i) - \min_i h_F(Z_i)$. Let ε_F^* be the smallest $\varepsilon > 0$ satisfying (9) with $h_{\max} = 1$, and define $C_F = \widehat{\operatorname{osc}}_n^F + 2M_F\varepsilon_F^*$.*

Then with probability at least $1-\delta$:

$$\operatorname{osc}_{\mathcal{K}_\alpha^F}(h_F) \;\le\; C_F \;=\; \widehat{\operatorname{osc}}_n^F + 2\,M_F\,\varepsilon_F^*. \tag{10}$$

The resulting α-HPD certified spectral-gap lower bound is

$$\underline{\gamma}_{F,\alpha} \;\ge\; \frac{2}{1+\exp(C_F)}. \tag{11}$$

This bounds the spectral gap of the independence MH chain restricted to $\mathcal{K}_\alpha^F$; for the full chain on D, $\underline{\gamma}_{F,\alpha}$ is a conservative certified lower bound whose tightness depends on the tail behaviour of h_F outside $\mathcal{K}_\alpha^F$. In all experiments we report quantile-core certificates (denoted QC $\underline{\gamma}$) at $\rho = 0.05$, following Hu [2026a].

Proof. Apply Hu [2026b, Theorem 5.1] with the quotient-metric covering (Section 3.4): the covering number is bounded by (7), the ball mass by $\pi_{\min}^F \cdot V_d \cdot \varepsilon^d$ (Assumption 3.7), and the interpolation error by $M_F \cdot d_G$ (Lemma 3.8). The Mengersen–Tweedie inequality [Mengersen and Tweedie, 1996] yields (11). ∎

4.2 Covering radius improvement

The folded certificate (10) has the same *form* as the unfolded one, but its three constituent quantities are more favourable. We begin with the covering radius.

Theorem 4.2 (Covering radius improvement)**.** *Let ε_U^* denote the smallest $\varepsilon > 0$ satisfying the unfolded covering condition*

$$\left(\frac{D_U}{\varepsilon} + 1\right)^d \cdot \left(1 - \pi_{\min}\,V_d\,\varepsilon^d\right)^{n_{\text{eff}}} \;\le\; \frac{\delta}{3}, \tag{12}$$

and let ε_F^* be defined by (9) *with $h_{\max} = 1$. Then $\varepsilon_F^* \le \varepsilon_U^*$.*

Proof. Write the left-hand side of the covering condition as $\Phi(D_K, \pi_{\min}, \varepsilon) = (D_K/\varepsilon + 1)^d (1 - \pi_{\min} V_d \varepsilon^d)^{n_{\text{eff}}}$. For any fixed $\varepsilon > 0$:

1. $D_F \le D_U$ (Corollary 3.3a), so the first factor does not increase;
2. $\pi_{\min}^F \ge s\,\pi_{\min} \ge \pi_{\min}$ (Corollary 3.3b), so $\pi_{\min}^F V_d \varepsilon^d \ge \pi_{\min} V_d \varepsilon^d$, which makes the base of the second factor smaller and hence the second factor does not increase.

Therefore $\Phi(D_F, \pi_{\min}^F, \varepsilon) \le \Phi(D_U, \pi_{\min}, \varepsilon)$ for all $\varepsilon > 0$. Since ε_U^* satisfies $\Phi(D_U, \pi_{\min}, \varepsilon_U^*) \le \delta/3$, we have $\Phi(D_F, \pi_{\min}^F, \varepsilon_U^*) \le \delta/3$, so by minimality $\varepsilon_F^* \le \varepsilon_U^*$. □

Remark 4.3 (Asymptotic scaling). Under simplifying approximations (dropping the $+1$ term and using $\log(1-p) \approx -p$), the ratio $\varepsilon_F^*/\varepsilon_U^*$ scales heuristically as $s^{-1/d}$, up to logarithmic factors. This explains the empirical observation that the covering-radius improvement is more pronounced in low dimensions (Section 5). However, the rigorous content of Theorem 4.2 is the inequality $\varepsilon_F^* \le \varepsilon_U^*$, which is independent of the cross-mode misspecification condition.

4.3 Certified gap improvement

The covering-radius improvement (Theorem 4.2) guarantees that the radius factor ε_F^* does not increase. The full product $M_F \varepsilon_F^*$ need not be smaller unless M_F is comparable to M_U, which is why the sufficient condition below keeps both terms explicit. The dominant source of improvement, however, is the empirical oscillation $\widehat{\text{osc}}_n^F$, which drops when the quotient proposal eliminates cross-mode residuals.

We now compare the folded and unfolded certificates. Let $T_U : \mathbb{R}^d \to \mathbb{R}^d$ be a normalising flow trained on the original (unfolded) target π, with proposal density $q_U = (T_U)_\# \mathcal{N}(0, I)$ and log-density ratio $h_U = \log \pi - \log q_U$. The unfolded certificate is $C_U = \widehat{\text{osc}}_n + 2M_U \varepsilon_U^*$, where $\widehat{\text{osc}}_n$, M_U, ε_U^* are defined analogously to their folded counterparts but using h_U and $\mathcal{K}_\alpha$.

To quantify the oscillation gap we introduce three conditions.

(M′) Relative cross-mode deficiency. For each off-mode component $\mathcal{K}_\alpha^{(j)}$, $j \ne 1$, there exists $\Delta_j > 0$ such that

$$\inf_{\theta \in \mathcal{K}_\alpha^{(j)}} h_U(\theta) \;\ge\; \sup_{\theta \in \mathcal{K}_\alpha^{(1)}} h_U(\theta) \;+\; \Delta_j. \tag{13}$$

(W) Folded within-mode fit. There exists $r_1 \ge 0$ such that

$$\text{osc}_{\mathcal{K}_\alpha^F}\bigl(\log(s\pi) - \log q_{T_F}\bigr) \;\le\; 2r_1. \tag{14}$$

(R) Folded-proposal residual.

$$r_F \;:=\; \sup_{z \in \mathcal{K}_\alpha^F} \bigl|\log q_F(z) - \log q_{T_F}(z)\bigr| \;<\; \infty. \tag{15}$$

Condition (M′) says the unfolded log-density ratio h_U is uniformly higher on every off-mode than on the primary mode: the unfolded flow under-covers those regions by at least a factor of e^{Δ_j}. This holds whenever the spectrally normalised RealNVP has insufficient capacity to represent multiple separated modes—a common situation in practice, as demonstrated by all our experiments (Section 5). Condition (W) bounds the within-mode oscillation of the folded flow's unnormalised log-density ratio on $\mathcal{K}_\alpha^F$. Condition (R) quantifies the contribution of off-mode orbit terms $q_{T_F}(g \cdot z)$ for $g \ne e$; for well-separated modes $r_F \approx 0$ since these terms are exponentially small.

Note that (M′) involves the unfolded flow T_U while (W) and (R) involve the folded flow T_F; this reflects the two-flow comparison at the heart of FolT-MCMC.

Lemma 4.4 (Oscillation gap)**.** *Under conditions* (M′), (W)*, and* (R)*, the empirical oscillations on the unfolded and folded spaces satisfy, with probability at least* $1-\delta'$*,*

$$\widehat{\mathrm{osc}}_n \;\geq\; \widehat{\mathrm{osc}}_n^F \;+\; \min_{j\neq 1}\Delta_j \;-\; 2r_1 - r_F \;-\; c_n, \tag{16}$$

where $c_n = O\big(\sqrt{\log(2/\delta')/n}\big)$ *accounts for the sampling error in both empirical oscillations.*

Proof. We bound $\widehat{\mathrm{osc}}_n$ from below and $\widehat{\mathrm{osc}}_n^F$ from above.

Lower bound on $\widehat{\mathrm{osc}}_n$*.* Condition (M′) gives, for any off-mode $j\neq 1$, $\inf_{\mathcal{K}_\alpha^{(j)}} h_U \geq \sup_{\mathcal{K}_\alpha^{(1)}} h_U + \Delta_j$. Therefore

$$\mathrm{osc}_{\mathcal{K}_\alpha}(h_U) \;\geq\; \inf_{\mathcal{K}_\alpha^{(j)}} h_U - \inf_{\mathcal{K}_\alpha^{(1)}} h_U \;\geq\; \sup_{\mathcal{K}_\alpha^{(1)}} h_U + \Delta_j - \inf_{\mathcal{K}_\alpha^{(1)}} h_U \;\geq\; \min_{j\neq 1}\Delta_j.$$

The empirical oscillation $\widehat{\mathrm{osc}}_n$ concentrates around $\mathrm{osc}_{\mathcal{K}_\alpha}(h_U)$ up to a term $O(\sqrt{\log/n})$.

Upper bound on $\widehat{\mathrm{osc}}_n^F$*.* For $z\in\mathcal{K}_\alpha^F = \mathcal{K}_\alpha^{(1)}\cap D$ (a.e.),

$$h_F(z) = \log(s\pi(z)) - \log q_F(z) = \underbrace{\log(s\pi(z)) - \log q_{T_F}(z)}_{\text{within-mode: osc}\leq 2r_1} \;-\; \log\Big(1 + \frac{\sum_{g\neq e} q_{T_F}(gz)}{q_{T_F}(z)}\Big).$$

The last term lies in $[-r_F, 0]$ by condition (R). Therefore $\mathrm{osc}_{\mathcal{K}_\alpha^F}(h_F) \leq 2r_1 + r_F$, and $\widehat{\mathrm{osc}}_n^F$ concentrates similarly.

Combining: $\widehat{\mathrm{osc}}_n - \widehat{\mathrm{osc}}_n^F \geq \min_j \Delta_j - 2r_1 - r_F - c_n$. □

We can now state the main comparison result.

Theorem 4.5 (Certified gap improvement)**.** *Adopt the setting of Theorem 4.1 and let* $C_U = \widehat{\mathrm{osc}}_n + 2M_U\varepsilon_U^*$ *be the unfolded certificate (with gradient constant* M_U *and covering radius* ε_U^**). Suppose conditions* (M′), (W), (R) *hold, and define*

$$\Delta \;=\; \min_{j\neq 1}\Delta_j - 2r_1 - r_F - c_n.$$

If

$$\Delta \;>\; 2\big(M_F\,\varepsilon_F^* \;-\; M_U\,\varepsilon_U^*\big)^+ \tag{17}$$

(where $(x)^+ = \max(x,0)$*), then with probability at least* $1-\delta_U-\delta_F-\delta'$*:*

$$C_F \;<\; C_U, \qquad \frac{\underline{\gamma}_F}{\underline{\gamma}_U} \;=\; \frac{1+\exp(C_U)}{1+\exp(C_F)} \;>\; 1, \tag{18}$$

where $\underline{\gamma}_F$ *and* $\underline{\gamma}_U$ *are the respective HPD-core certified lower bounds.*

Proof. Write

$$\begin{aligned} C_U - C_F &= \big(\widehat{\mathrm{osc}}_n - \widehat{\mathrm{osc}}_n^F\big) + 2\big(M_U\varepsilon_U^* - M_F\varepsilon_F^*\big) \\ &\geq \Delta + 2\big(M_U\varepsilon_U^* - M_F\varepsilon_F^*\big) \qquad \text{(Lemma 4.4)}. \end{aligned}$$

If $M_F\varepsilon_F^* \leq M_U\varepsilon_U^*$, both terms are non-negative and $C_U - C_F \geq \Delta > 0$. If $M_F\varepsilon_F^* > M_U\varepsilon_U^*$, condition (17) gives $\Delta > 2(M_F\varepsilon_F^* - M_U\varepsilon_U^*)$, so again $C_U - C_F > 0$. The certified lower-bound comparison follows from the monotonicity of $C\mapsto 2/(1+e^C)$. The confidence level combines the three events via a union bound: the folded certificate (prob $\geq 1-\delta_F$), the unfolded certificate ($\geq 1-\delta_U$), and the oscillation-gap concentration ($\geq 1-\delta'$). □

4.4 When does folding help?

Theorems 4.2 and 4.5 identify two complementary improvement mechanisms: a covering-radius reduction that holds independently of the cross-mode deficiency condition ($\varepsilon_F^* \le \varepsilon_U^*$, under Assumption 3.7) and an oscillation reduction that holds under condition (M′).

The role of (M′). Condition (M′) is *not* an assumption about the target; it is a condition on the *unfolded flow* T_U. It asserts that the trained unfolded transport map fails to cover at least one off-mode. This holds naturally for spectrally normalised RealNVP with scale clipping, whose limited expressiveness makes faithful representation of multiple separated modes difficult—especially in higher dimensions, where the coupling architecture must mediate inter-coordinate information through the conditioner network. In all our experiments (Section 5), condition (M′) is strongly satisfied, with $\Delta/C_U \in [0.75, 0.87]$.

When the flow *is* expressive enough to represent all modes faithfully, $\Delta \approx 0$ and the oscillation gap vanishes; the improvement then comes solely from the covering-radius reduction (Theorem 4.2), which is modest—heuristically $O(s^{-1/d})$ (Remark 4.3).

Fold boundary placement. The main-text results (Theorems 4.1–4.5) are stated under Assumption 3.7, i.e. the HPD set does not touch the fold boundary. The boundary-touching case is handled by the stabiliser-corrected covering condition (9) and is detailed in Supplement A. When the fold boundary passes through a high-density region—as in our banana diagnostic experiment (Section 5.2)—the full oscillation can increase due to boundary residuals, even while the quantile-core certificate improves. This provides a practical design principle: *the fold boundary should lie in a low-density region separating the symmetric modes.* For label-switching mixtures, the sorted-chamber boundary is automatically in the valley between modes whenever the components are well separated.

Dimension dependence. A key empirical finding is that $\underline{\gamma}_F$ is empirically nearly dimension-free: it stays in $[0.90, 0.94]$ across $d = 2$ to 20 for Gaussian mixtures, while $\underline{\gamma}_U$ collapses from 0.40 to 0.016 (Section 5.3). The theoretical explanation is that all three terms in C_F remain small as d grows: $\widehat{\mathrm{osc}}_n^F$ reflects only the single-mode transport residual, M_F is controlled by spectral normalisation, and ε_F^* benefits from the s-fold density floor. In contrast, C_U is dominated by the cross-mode oscillation, which grows with d because the coupling-layer architecture struggles increasingly to generate multimodal structure in high dimensions.

Mode count vs. dimension. The label-switching experiments (Section 5.4) reveal that the unfolded certificate degrades primarily with the number of modes $s = m!$, not with the dimension d. Comparing $m = 3, p = 2$ ($d = 6$, $s = 6$) with $m = 3, p = 4$ ($d = 12$, $s = 6$): doubling the dimension while keeping s fixed reduces $\underline{\gamma}_U$ by only a factor of 2, whereas increasing s from 6 to 24 (at fixed block size) reduces it by a factor of 15. The folded certificate is insensitive to both.

5 Experiments

We evaluate FolT-MCMC on three synthetic target families designed to isolate the mechanisms predicted by the theory, followed by a comparison with the random permutation sampler [Frühwirth-Schnatter, 2001]. The first experiment is a diagnostic stress test showing that folding can be harmful when the fold boundary intersects high-density mass. The second evaluates dimension scaling for a well-separated reflection-symmetric Gaussian mixture. The third evaluates permutation folding for label-switching mixtures. The fourth compares FolT-MCMC with the standard label-switching baseline. Across experiments, we report quantile-core oscillation certificates, certified lower bounds, NLL, acceptance, and ESS. All experiments use the same architecture and training protocol; the only variable across conditions is whether folding is applied.

5.1 Setup

Architecture. Both the folded and unfolded flows are spectrally normalised RealNVP [Dinh et al., 2017, Miyato et al., 2018] with tanh activations and scale clipping at $c = 0.7$, following the LCNF protocol [Hu, 2026b]. Layer count and hidden dimension scale with d: $(L, h) = (8, 64)$ for $d \leq 4$; $(10, 128)$ for $d = 5$–6; $(12, 128)$ for $d = 8$–12; $(16, 256)$ for $d = 20$. The folded and unfolded pipelines share identical architecture for each d.

Training. Negative log-likelihood with annealed oscillation and gradient regularisation (FolT-OG-Anneal), as in CerT-MCMC [Hu, 2026a]. Training uses Adam with learning rate 10^{-3}, batch size 512, and 2 000 epochs for $d \leq 6$ (increasing to 3 500 for $d \geq 8$). Training samples are drawn from the exact sampler of each target (folded samples projected onto D).

Certification. Quantile-core certificates at $\rho = 0.05$ (primary) and $\rho = 0.025$ (secondary), with $n = 20\,000$ independent certification samples and DKW-corrected confidence $\delta = 0.05$, following the CerT-MCMC v2 protocol.

MCMC. Independence Metropolis–Hastings with 4 chains of length 5 000 (burn-in 1 000). Effective sample size (ESS) estimated via batch means.

Hardware and code. NVIDIA RTX 4090 GPU; code at https://github.com/junhu22/FolT-MCMC.

5.2 Diagnostic: fold boundary location

Purpose. To demonstrate that the fold boundary's position relative to the target's density mass is a critical design variable, and that folding can *hurt* the full oscillation bound when the boundary passes through a high-density region.

Target. Asymmetric double banana in $d = 2$: a mixture of two banana-shaped modes with curvatures $a = 1.0$ and $b = 0.5$, related by reflection $\theta_1 \mapsto -\theta_1$. Because $a \neq b$, the target is not strictly $\mathbb{Z}_2$-invariant; this experiment intentionally stresses the method outside the exact group-invariant setting of Section 3. The fold boundary $\{\theta_1 = 0\}$ passes through the high-density region near the origin where both modes overlap.

Results. Table 1 summarises the comparison.

The full oscillation *increases* under folding (58.4 vs. 30.7), because the smooth flow cannot produce the hard truncation at $\theta_1 = 0$ where the target density is high, creating a boundary residual spike. Nevertheless, the quantile-core certified lower bound still improves: QC $\underline{\gamma}$ rises from 0.272 to 0.320 at $\rho = 0.05$ and from 0.064 to 0.135 at $\rho = 0.025$, because the ρ-trimming excludes the boundary spike. The NLL drops by 23% (4.34 → 3.34), confirming that the flow fits the single folded mode more easily.

Table 1: Banana diagnostic ($d = 2$, $s = 2$). The fold boundary passes through a high-density region.

Metric	Unfolded	Folded
Full oscillation	30.7	58.4
QC oscillation ($\rho = 0.05$)	1.38	1.10
QC $\underline{\gamma}$ ($\rho = 0.05$)	0.272	0.320
QC $\underline{\gamma}$ ($\rho = 0.025$)	0.064	0.135
Final NLL	4.34	3.34
ESS / sample	0.048	0.042
Acceptance rate	0.604	0.594

Table 2: Dimension scaling for Gaussian mixture ($s = 2$, $\rho = 0.05$).

d	QC $\underline{\gamma}_U$	QC $\underline{\gamma}_F$	Ratio	QC osc$_U$	QC osc$_F$
2	0.402	0.936	2.3×	1.38	0.13
5	0.094	0.941	10.0×	3.01	0.12
10	0.016	0.922	59.2×	4.85	0.16
20	0.016	0.902	57.0×	4.83	0.20

This experiment validates the design principle of Section 4.4: *folding helps when the fold boundary lies in a low-density valley, not when it cuts through the target's mass.* The subsequent experiments use targets satisfying this condition.

5.3 Dimension scaling: well-separated mixture

Purpose. To test whether the folded certificate is robust to increasing dimension, in contrast to the unfolded certificate.

Target. Symmetric two-component Gaussian mixture, $\pi = \frac{1}{2}\mathcal{N}(\mu_1, I_d) + \frac{1}{2}\mathcal{N}(\mu_2, I_d)$, with $\mu_1 = (3, 0, \ldots, 0)$, $\mu_2 = -\mu_1$ (separation $\delta = 6$). Reflection fold with $G = \mathbb{Z}_2$, $s = 2$. Dimensions $d \in \{2, 5, 10, 20\}$. The fold boundary $\{\theta_1 = 0\}$ sits in the inter-mode valley where the density relative to a single-component peak is $\exp(-9/2) \approx 0.011$.

Results. Table 2 presents the key comparison.

The folded quantile-core certificate QC $\underline{\gamma}_F$ remains in $[0.90, 0.94]$ across $d = 2$ to 20, while the unfolded certificate collapses from 0.40 to 0.016—a 59-fold gap by $d = 10$. The quantile-core oscillation bound tells the same story: folded QC oscillation stays below 0.20 at all dimensions, while the unfolded QC oscillation grows from 1.38 to 4.85.

At $d = 20$, the folded NLL of 28.48 is within 0.1 of the entropy of a d-dimensional standard Gaussian, $\frac{d}{2}\log(2\pi e) \approx 28.4$, indicating that the folded transport achieves a near-optimal fit. The unfolded flow, by contrast, must represent a bimodal target in 20 dimensions using a coupling architecture that can only influence each coordinate conditionally—a task at which it demonstrably fails.

5.4 Label-switching scaling

Purpose. To verify that permutation folding gives the same strong improvement as reflection folding, and to disentangle the effects of mode count s and dimension d on the unfolded certificate.

Table 3: Label-switching scaling (quantile-core, $\rho = 0.05$).

Config	d	s	QC $\underline{\gamma}_U$	QC $\underline{\gamma}_F$	Ratio	QC osc_U	QC osc_F
m=2, p=2	4	2	0.366	0.820	2.2×	1.50	0.37
m=3, p=2	6	6	0.066	0.685	10.4×	3.38	0.65
m=3, p=4	12	6	0.036	0.683	19.1×	4.01	0.66
m=4, p=2	8	24	0.004	0.624	145×	6.14	0.79

Target. Label-switching Gaussian mixtures with m components, each a block of p parameters (representing, e.g., frequency and damping of a structural mode), total dimension $d = mp$. Component centres are spaced by 3 standard deviations along the sort coordinate. Permutation fold with $G = S_m$, $s = m!$. Four configurations are tested (Table 3).

Results. The folded quantile-core certificate QC $\underline{\gamma}_F$ stays in $[0.62, 0.82]$ across all configurations, while the unfolded certificate collapses from 0.366 to 0.004 as the number of equivalent modes grows from $s = 2$ to $s = 24$. The improvement ratio reaches 145× at $m = 4$.

Disentangling mode count from dimension. Comparing $m = 3, p = 2$ ($d = 6$, $s = 6$, QC $\underline{\gamma}_U = 0.066$) with $m = 3, p = 4$ ($d = 12$, $s = 6$, QC $\underline{\gamma}_U = 0.036$): doubling the dimension at fixed s reduces QC $\underline{\gamma}_U$ by a factor of ≈ 2. Comparing $m = 3, p = 2$ ($s = 6$) with $m = 4, p = 2$ ($s = 24$): increasing s by 4× at comparable d reduces QC $\underline{\gamma}_U$ by a factor of ≈ 15. In this controlled comparison, the unfolded certificate appears more sensitive to the number of symmetric copies than to the ambient dimension—consistent with the theoretical prediction of Section 4.4. The folded certificate varies much less across both changes (QC $\underline{\gamma}_F \in [0.68, 0.68]$ for the two $s = 6$ configurations despite d doubling).

Convergence at $m = 4$. With $s = 24$ equivalent modes, the unfolded flow trains stably (no divergence or numerical issues) but fails representationally: NLL = 15.0, acceptance rate = 0.41, full-oscillation certificate $\gamma_{\text{full}} \approx 5 \times 10^{-10}$ (completely vacuous). Scatter plots confirm that the unfolded proposal spreads mass across the 24 symmetric chambers but yields a highly uneven density ratio, leading to low acceptance and a vacuous certificate. The folded chain, by contrast, samples efficiently in the sorted chamber (acceptance = 0.90, ESS/sample = 0.85).

5.5 Comparison with random permutation sampler

Purpose. To compare FolT-MCMC with the random permutation sampler (RPS) of Frühwirth-Schnatter [2001], the most widely used label-switching mitigation in Bayesian mixture modelling.

Setup. All three methods use the $m = 3$, $p = 2$ label-switching target ($d = 6$, $s = 6$) from Section 5.4. The unfolded IMH and RPS share the same trained flow as proposal; RPS additionally applies a random component-label permutation after each MH accept/reject step. FolT-MCMC uses its own folded flow in the sorted chamber. Chains: $4 \times 10\,000$ steps, burn-in $2\,000$. To ensure a fair mixing comparison, we report ESS after projecting all samples into the sorted fundamental domain.

Results. The sorted ESS per sample is comparable across all three methods (0.48–0.61), well within the variance of the batch-means estimator. This is expected: all three use a global independence proposal from a well-trained flow that already covers all six modes, so no chain gets stuck

Table 4: Baseline comparison ($m = 3$, $d = 6$, $s = 6$). Sorted ESS is computed after projecting all samples into the sorted chamber.

Method	Sorted ESS / sample	Acceptance	Time (s)	QC $\underline{\gamma}$
Unfolded IMH	0.607	0.622	0.07	0.066
FolT-MCMC	0.540	0.881	0.07	0.685
RPS	0.482	0.612	682.8	N/A

in a single mode—the setting where RPS was designed to help [Frühwirth-Schnatter, 2001]. RPS incurs a large wall-clock overhead in our implementation (682.8 s vs. 0.07 s), because the per-sample permutation step is not batched.

The real differentiator is the certified guarantee. FolT-MCMC's certificate QC $\underline{\gamma} = 0.685$ predicts ESS/$n \approx \gamma/(2-\gamma) = 0.52$, matching the empirical 0.54—a tight certificate. The unfolded certificate QC $\underline{\gamma} = 0.066$ predicts ESS/$n \approx 0.034$, far below the empirical 0.61: the certificate is valid but loose, reflecting worst-case cross-mode oscillation that did not materialise on this particular run. For RPS, we do not have an LCNF/Mengersen–Tweedie certificate: its composite IMH-plus-permutation kernel does not fit the independence-proposal minorisation structure underlying our certification framework.

This comparison underscores the core thesis of FolT-MCMC: *folding does not primarily improve raw mixing speed; it improves the certifiability of the chain*, closing the gap between the provable lower bound and the empirical performance.

6 Application: closely-spaced structural modes

We apply FolT-MCMC to Bayesian modal identification of a supertall building instrumented during Typhoon Mangkhut (2018). Three closely-spaced modes in the 0.7–1.1 Hz band create an S_3 label-switching problem ($s = 6$) that renders the unfolded certificate vacuous.

6.1 Data and model

Data. Accelerometer recordings from 15 usable channels at 32 Hz. We select a single 30-minute window near peak typhoon intensity (band signal-to-noise ratio ≈ 18 dB) and compute the trace of the 15×15 cross-power spectral density matrix over 0.7–1.1 Hz (51 frequency ordinates), capturing the total vibrational power across all channels. Building identity and location are anonymised per the owner's request.

Model. A simplified Whittle likelihood with three single-degree-of-freedom Lorentzian peaks plus a flat noise floor. Each mode j has two parameters (natural frequency f_j, damping ratio ξ_j), giving $d = 6$ total. The overall amplitude is profiled out in closed form (conditional MLE), preserving exact S_3 symmetry. Uniform priors: $f_j \in [0.7, 1.1]$ Hz, $\xi_j \in [0.001, 0.05]$.

This is deliberately simpler than a full multi-output Bayesian operational modal analysis: the goal is to exhibit the label-switching phenomenon and demonstrate FolT-MCMC's certifiability advantage, not to compete on estimation accuracy.

Folding. Permutation fold with $G = S_3$, $s = 6$. The fundamental domain is the sorted chamber $D = \{f_1 \leq f_2 \leq f_3\}$. Parameters are block-whitened (one shared mean and standard deviation for the frequency slots, one for the damping slots) to bring all coordinates to unit scale while preserving the commutativity of whitening and sorting.

Table 5: Structural modal identification ($d = 6$, $s = 6$, Typhoon Mangkhut window).

Method	QC $\underline{\gamma}$ (ρ=0.05)	Acceptance	ESS / sample	Label-switch rate
Unfolded	9.9×10^{-6} (vacuous)	0.204	0.059	0.170
FolT-MCMC	0.0018	0.371	0.080	0.000

6.2 Results

Training uses 50 000 pre-samples from a long-run random-walk Metropolis chain, folded onto D. Architecture and certification protocol follow Section 5.1.

Label switching. The unfolded chain exhibits dense label switching, with 17% of steps swapping the frequency assignment among the three modes (Table 5). The FolT-MCMC chain in sorted space shows zero label switches: the sorted frequencies remain cleanly separated throughout the run.

Certified guarantee. The unfolded quantile-core certificate is QC $\underline{\gamma} \approx 10^{-5}$—completely vacuous, as the six permutation modes inflate the oscillation bound beyond any useful range. FolT-MCMC provides a non-vacuous quantile-core certificate ($\underline{\gamma} = 0.0018$), a 180× improvement. The absolute value is modest, reflecting the genuine difficulty of closely-spaced modes: the near-degenerate boundary $f_i \approx f_j$ in the sorted chamber creates heavy-tailed density ratios that resist tight certification. Nevertheless, the qualitative transition from *no usable certificate* to *a certificate* is the operationally meaningful distinction.

Posterior estimates. The posterior median frequencies are $\hat{f}_1 = 0.804$, $\hat{f}_2 = 0.838$, $\hat{f}_3 = 0.873$ Hz, recovering the closely-spaced cluster structure. These values are approximately 0.05 Hz below the nominal multi-output estimates from the full Bayesian OMA of the same building [Lam et al., 2019], a systematic offset attributable to the single-output trace-spectrum model, which under-weights the transverse mode ($\approx$0.93 Hz) relative to the translational modes. The demonstration targets the sampler's certifiability, not estimation accuracy; the full AD-BOMA model is developed separately [Hu and Lam, 2026].

7 Discussion

7.1 Design principles

The experiments point to a simple but consequential design rule: *the fold boundary should lie in a low-density valley between the symmetric modes*. When this condition holds (Gaussian mixture, label-switching targets), folding yields large certified gains across all metrics. When it is violated (banana diagnostic with fold boundary through the density ridge), the full oscillation bound can increase, though the quantile-core certificate still improves modestly.

For the most common use cases—label-switching in mixture models and closely-spaced structural modes—this condition is satisfied automatically. The sorted-chamber boundary $\{\theta_1^{(j)} = \theta_1^{(j+1)}\}$ lies at the midpoint between adjacent mode centres, which is the lowest-density region when components are separated by more than a few standard deviations.

7.2 Relation to existing work

Equivariant normalising flows. Köhler et al. [2020] and Rezende et al. [2019] construct flows satisfying $T(g \cdot z) = g \cdot T(z)$, encoding the symmetry into the architecture. This reduces the number of learnable parameters and ensures that the flow respects the group structure, but does not reduce the number of modes: the equivariant flow must still represent all $|G|$ copies. FolT-MCMC takes the dual approach of collapsing the $|G|$ copies into one via the quotient map. The two strategies are complementary: one could train an equivariant flow and then fold, combining parameter efficiency with mode reduction.

Post-hoc relabelling. Methods such as pivotal reordering [Stephens, 2000], equivalence-class representatives [Papastamoulis and Iliopoulos, 2010], and random permutation samplers [Frühwirth-Schnatter, 2001] operate after MCMC has been run. They improve the interpretability of the samples but do not address the mixing difficulty of the underlying chain. In particular, no post-hoc method provides a certified spectral-gap bound. FolT-MCMC resolves the mixing problem at source by eliminating the symmetric modes before sampling, and inherits certification from the LCNF framework.

Tempering and replica exchange. Parallel tempering [Geyer, 1991, Earl and Deem, 2005] overcomes multimodality by introducing heated replicas that can cross energy barriers. This is a general-purpose strategy that does not exploit symmetry. FolT-MCMC exploits the *structure* of the multimodality (it arises from a known group action) to remove it geometrically, without the overhead of maintaining multiple temperature levels.

7.3 Limitations and future work

Exact symmetry. The current framework requires the target to be exactly G-invariant. The simplified exchangeable model in Section 6 preserves exact label symmetry by construction (identical Lorentzian parameterisation and profiled amplitude). In more realistic Bayesian operational modal analysis models, however, mode-specific priors, spatial mode-shape information, or physical ordering constraints may break this symmetry approximately. Extending FolT-MCMC to ε-approximate group invariance, where $\|\pi(g\theta) - \pi(\theta)\|$ is small but nonzero, is a natural direction. We expect the quotient certificate to degrade gracefully with the symmetry-breaking magnitude, but a formal perturbation bound remains to be established.

Unknown symmetry groups. FolT-MCMC presupposes knowledge of G and of a fundamental domain D. For the applications considered here (mixture models, structural identification), the symmetry is known by construction. When the symmetry is latent or only partially known, a *learned folding map* could be trained jointly with the transport flow, analogous to recent work on learning group structure from data [Dehmamy et al., 2021].

Continuous symmetries. Finite groups cover label switching and discrete reflections, but some inference problems have continuous symmetries (rotation invariance, gauge symmetry in physics). Extending the quotient construction to compact Lie groups would require replacing the finite orbit sum in the quotient proposal with a Haar-measure integral, and the covering theory with manifold covering bounds.

Scalability. The orbit sum in the quotient proposal has $|G|$ terms. For S_m with moderate m ($m \leq 5$, $|G| \leq 120$), this is inexpensive. For larger groups, truncation or importance sampling over the orbit may be needed.

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A Complete proofs

A.1 Proof of Proposition 3.2 (HPD identity)

Proof. Let $U(\theta) = -\log \pi(\theta) + \log Z_\pi$ denote the potential of π, where Z_π is the normalising constant. The folded potential is $U_F(z) = -\log \pi_F(z) + \log Z_{\pi_F} = -\log(s\pi(z)) + \log 1 = U(z) - \log s$, using $Z_{\pi_F} = 1$ since π_F integrates to one on D.

Part (i). We show that $U(Z_F)$ under $Z_F \sim \pi_F$ has the same distribution as $U(\Theta)$ under $\Theta \sim \pi$. For any Borel function $\varphi\colon \mathbb{R} \to [0, \infty)$, the G-invariance of π and fundamental-domain decomposition give

$$\begin{aligned}
\mathbb{E}_{\Theta\sim\pi}[\varphi(U(\Theta))] &= \int_{\mathbb{R}^d} \varphi(U(\theta))\, \pi(\theta)\, \mathrm{d}\theta \\
&= \sum_{g\in G} \int_{gD} \varphi(U(\theta))\, \pi(\theta)\, \mathrm{d}\theta \\
&= \sum_{g\in G} \int_{D} \varphi(U(g\cdot z))\, \pi(g\cdot z)\, \mathrm{d}z \qquad (\text{substituting } \theta = g\cdot z,\ |\det g| = 1) \\
&= s \int_{D} \varphi(U(z))\, \pi(z)\, \mathrm{d}z \qquad (\text{since } U(g\cdot z) = U(z) \text{ and } \pi(g\cdot z) = \pi(z)) \\
&= \int_{D} \varphi(U(z))\, \pi_F(z)\, \mathrm{d}z = \mathbb{E}_{Z_F\sim\pi_F}[\varphi(U(Z_F))]. \qquad (19)
\end{aligned}$$

Since (19) holds for all non-negative Borel φ, the distributions of $U(\Theta)$ and $U(Z_F)$ coincide. In particular, their $(1-\alpha)$-quantiles are equal: the $(1-\alpha)$-quantile of $U(Z_F)$ is u_α. Since $U_F(Z_F) = U(Z_F) - \log s$, the $(1-\alpha)$-quantile of $U_F(Z_F)$ is $u_\alpha - \log s =: u_\alpha^F$.

Part (ii).

$$\begin{aligned}
\mathcal{K}_\alpha^F &= \{z \in D : U_F(z) \le u_\alpha^F\} = \{z \in D : U(z) - \log s \le u_\alpha - \log s\} \\
&= \{z \in D : U(z) \le u_\alpha\} = \mathcal{K}_\alpha \cap D.
\end{aligned}$$

□

A.2 Proof of Corollary 3.3

Proof. **Part (a).** Since $\mathcal{K}_\alpha^F = \mathcal{K}_\alpha \cap D \subseteq \mathcal{K}_\alpha$, we have $\operatorname{diam}(\mathcal{K}_\alpha^F) \le \operatorname{diam}(\mathcal{K}_\alpha)$.

Part (b).

$$\pi_{\min}^F = \inf_{z\in\mathcal{K}_\alpha^F} \pi_F(z) = s \cdot \inf_{z\in\mathcal{K}_\alpha\cap D} \pi(z).$$

Since $\mathcal{K}_\alpha \cap D \subseteq \mathcal{K}_\alpha$, $\inf_{\mathcal{K}_\alpha\cap D} \pi \ge \inf_{\mathcal{K}_\alpha} \pi = \pi_{\min}$, so $\pi_{\min}^F \ge s\,\pi_{\min}$.

When π is G-invariant and all orbits in $\mathcal{K}_\alpha$ have a representative in D (which holds by definition of a fundamental domain), the G-invariance implies that $\operatorname{ess\,inf}_{\mathcal{K}_\alpha\cap D} \pi = \operatorname{ess\,inf}_{\mathcal{K}_\alpha} \pi$, giving $\pi_{\min}^F = s\,\pi_{\min}$ in the essential-infimum sense. □

A.3 Proof of Lemma 3.8 (quotient Lipschitz constant)

Proof. The extended log-density ratio on $\mathbb{R}^d$ is

$$h_F(\theta) = \log\bigl(s\,\pi(\theta)\bigr) - \log\Bigl(\sum_{g\in G} q_{T_F}(g\cdot\theta)\Bigr).$$

Both π and $\theta \mapsto \sum_g q_{T_F}(g\cdot\theta)$ are G-invariant, so $h_F(g\cdot\theta) = h_F(\theta)$ for all $g \in G$ and $\theta \in \mathbb{R}^d$.

Fix $x, y \in \mathcal{K}_\alpha^F$ and an arbitrary $g \in G$. The segment from x to $g \cdot y$ has length $\|x - g \cdot y\|$. Since $d_G(x, y) \leq \|x - g \cdot y\|$ and $\mathcal{K}_\alpha^F$ has diameter bounded by $\mathrm{diam}(\mathcal{K}_\alpha)$, the segment lies within the ε_F^*-neighbourhood $\mathcal{K}_{\alpha,\varepsilon}^{G,+} = \{w \in \mathbb{R}^d : d_G(w, \mathcal{K}_\alpha^F) \leq \varepsilon_F^*\}$ provided $\|x - g \cdot y\| \leq \mathrm{diam}(\mathcal{K}_\alpha^F) + \varepsilon_F^*$, which holds for the minimising g. By the mean value theorem,

$$|h_F(x) - h_F(g \cdot y)| \;\leq\; \sup_{w \in \mathcal{K}_{\alpha,\varepsilon}^{G,+}} \|\nabla h_F(w)\| \;\cdot\; \|x - g \cdot y\| \;=\; M_F \cdot \|x - g \cdot y\|.$$

Using $h_F(g \cdot y) = h_F(y)$ (by G-invariance) and minimising over $g \in G$:

$$|h_F(x) - h_F(y)| = |h_F(x) - h_F(g \cdot y)| \leq M_F \cdot \min_{g \in G} \|x - g \cdot y\| = M_F \cdot d_G(x, y). \qquad \square$$

A.4 Proof of Theorem 4.1 (FolT-MCMC certificate)

Proof. The proof instantiates Hu [2026b, Theorem 5.1] on the quotient setting. We verify that the three inputs required by that theorem are available.

Input 1: Covering number bound. Since $d_G(x, y) \leq \|x - y\|$, any Euclidean ε-cover of $\mathcal{K}_\alpha^F$ is also a d_G-ε-cover. The standard volumetric argument gives

$$\mathcal{N}_G(\mathcal{K}_\alpha^F, \varepsilon) \leq \mathcal{N}_E(\mathcal{K}_\alpha^F, \varepsilon) \leq \Big(\frac{2D_F}{\varepsilon} + 1\Big)^d,$$

where $D_F = \mathrm{diam}(\mathcal{K}_\alpha^F)$.

Input 2: Ball-mass lower bound. Under Assumption 3.7, $|\mathrm{Stab}(z)| = 1$ for all $z \in \mathcal{K}_\alpha^F$. For such z, the quotient ball $B_G(z, \varepsilon) = \{z' : d_G(z, z') \leq \varepsilon\}$ corresponds in $\mathbb{R}^d$ to the union $\bigcup_{g \in G}(B(g \cdot z, \varepsilon) \cap g \cdot D)$. The s pieces tile a full Euclidean ball (up to the measure-zero boundaries $\partial(g \cdot D)$), so $\mathrm{Vol}(B_G(z, \varepsilon)) = V_d\, \varepsilon^d$. Therefore

$$\pi_F(B_G(z, \varepsilon) \cap D) \;\geq\; \pi_{\min}^F \cdot V_d \cdot \varepsilon^d.$$

Input 3: Lipschitz interpolation. Lemma 3.8 gives $|h_F(x) - h_F(y)| \leq M_F \cdot d_G(x, y)$.

Substituting these three inputs into Hu [2026b, Theorem 5.1], the covering condition

$$\Big(\frac{D_F}{\varepsilon} + 1\Big)^d \big(1 - \pi_{\min}^F\, V_d\, \varepsilon^d\big)^{n_{\text{eff}}} \leq \frac{\delta}{3}$$

guarantees that with probability $\geq 1 - \delta$, every point of $\mathcal{K}_\alpha^F$ lies within d_G-distance ε_F^* of some certification sample. The oscillation bound follows:

$$\mathrm{osc}_{\mathcal{K}_\alpha^F}(h_F) \leq \widehat{\mathrm{osc}}_n^F + 2M_F \varepsilon_F^*.$$

Applying the Mengersen–Tweedie inequality [Mengersen and Tweedie, 1996] to the $\mathcal{K}_\alpha^F$-restricted oscillation yields the HPD-core certified lower bound $\underline{\gamma}_{F,\alpha} \geq 2/(1 + \exp(\widehat{\mathrm{osc}}_n^F + 2M_F \varepsilon_F^*))$. $\square$

A.5 Proof of Theorem 4.2 (covering radius improvement)

Proof. Write the left-hand side of the covering condition as

$$\Phi(D_K,\, \pi_{\min},\, \varepsilon) \;=\; \Big(\frac{D_K}{\varepsilon} + 1\Big)^d \big(1 - \pi_{\min}\, V_d\, \varepsilon^d\big)^{n_{\text{eff}}}.$$

Monotonicity in D_K. For fixed ε and $\pi_{\min}$, Φ is increasing in D_K (the first factor $(D_K/\varepsilon + 1)^d$ increases).

Monotonicity in $\pi_{\min}$. For fixed ε and D_K, Φ is decreasing in $\pi_{\min}$ (the base $1-\pi_{\min} V_d\,\varepsilon^d$ decreases, and since $n_{\text{eff}}>0$ and the base is in $(0,1)$, the second factor decreases).

By Corollary 3.3: (a) $D_F=\operatorname{diam}(\mathcal{K}_\alpha^F)\le D_U=\operatorname{diam}(\mathcal{K}_\alpha)$; (b) $\pi_{\min}^F\ge s\,\pi_{\min}\ge\pi_{\min}$. Therefore, for every $\varepsilon>0$,

$$\Phi(D_F,\,\pi_{\min}^F,\,\varepsilon)\ \le\ \Phi(D_U,\,\pi_{\min},\,\varepsilon).$$

Since ε_U^* satisfies $\Phi(D_U,\pi_{\min},\varepsilon_U^*)\le\delta/3$, we have $\Phi(D_F,\pi_{\min}^F,\varepsilon_U^*)\le\delta/3$. By the minimality of ε_F^* (smallest ε satisfying the folded covering condition), $\varepsilon_F^*\le\varepsilon_U^*$. □

A.6 Proof of Lemma 4.4 (oscillation gap)

Proof. Let $h_U=\log\pi-\log q_U$ (unfolded log-density ratio) and $h_F=\log\pi_F-\log q_F$ (folded log-density ratio). We bound $\widehat{\operatorname{osc}}_n$ from below and $\widehat{\operatorname{osc}}_n^F$ from above.

Step 1: Lower bound on $\operatorname{osc}_{\mathcal{K}_\alpha}(h_U)$. The HPD set decomposes into connected components $\mathcal{K}_\alpha=\bigcup_{j=1}^k\mathcal{K}_\alpha^{(j)}$ (one per mode). Condition (M$'$) gives, for each $j\ne1$,

$$\inf_{\theta\in\mathcal{K}_\alpha^{(j)}}h_U(\theta)\ \ge\ \sup_{\theta\in\mathcal{K}_\alpha^{(1)}}h_U(\theta)+\Delta_j.$$

Therefore

$$\begin{aligned}
\operatorname{osc}_{\mathcal{K}_\alpha}(h_U)&\ge\sup_{\mathcal{K}_\alpha^{(j)}}h_U-\inf_{\mathcal{K}_\alpha^{(1)}}h_U\\
&\ge\inf_{\mathcal{K}_\alpha^{(j)}}h_U-\inf_{\mathcal{K}_\alpha^{(1)}}h_U\\
&\ge\Big(\sup_{\mathcal{K}_\alpha^{(1)}}h_U+\Delta_j\Big)-\inf_{\mathcal{K}_\alpha^{(1)}}h_U\ \ge\ \Delta_j.
\end{aligned}$$

Taking the minimum over $j\ne1$: $\operatorname{osc}_{\mathcal{K}_\alpha}(h_U)\ge\min_{j\ne1}\Delta_j$.

The empirical oscillation $\widehat{\operatorname{osc}}_n$ is computed from n i.i.d. samples from π. By standard extreme-value concentration (see, e.g., Hu 2026b, Lemma 5.3),

$$\mathbb{P}\big(\operatorname{osc}_{\mathcal{K}_\alpha}(h_U)-\widehat{\operatorname{osc}}_n>t\big)\le2\exp(-2nt^2/R^2)$$

for a range parameter R. Setting the right-hand side to $\delta'/2$ gives $\widehat{\operatorname{osc}}_n\ge\operatorname{osc}_{\mathcal{K}_\alpha}(h_U)-c_n'$ with probability $\ge1-\delta'/2$, where $c_n'=O(\sqrt{\log(2/\delta')/n})$.

Step 2: Upper bound on $\operatorname{osc}_{\mathcal{K}_\alpha^F}(h_F)$. For $z\in\mathcal{K}_\alpha^F=\mathcal{K}_\alpha^{(1)}\cap D$ (a.e. by Proposition 3.2),

$$h_F(z)=\log(s\,\pi(z))-\log q_F(z)=\underbrace{\big[\log(s\,\pi(z))-\log q_{T_F}(z)\big]}_{=:\tilde h(z)}\ -\ \underbrace{\log\!\Big(1+\frac{\sum_{g\ne e}q_{T_F}(g\cdot z)}{q_{T_F}(z)}\Big)}_{=:\rho(z)}.$$

By condition (W), $\operatorname{osc}_{\mathcal{K}_\alpha^F}(\tilde h)\le2r_1$.

By condition (R), $|\rho(z)|\le r_F$ for all $z\in\mathcal{K}_\alpha^F$, so $\operatorname{osc}_{\mathcal{K}_\alpha^F}(\rho)\le r_F$.

Since $h_F=\tilde h-\rho$,

$$\operatorname{osc}_{\mathcal{K}_\alpha^F}(h_F)\le\operatorname{osc}_{\mathcal{K}_\alpha^F}(\tilde h)+\operatorname{osc}_{\mathcal{K}_\alpha^F}(\rho)\le2r_1+r_F.$$

By the same concentration argument, $\widehat{\operatorname{osc}}_n^F\le\operatorname{osc}_{\mathcal{K}_\alpha^F}(h_F)+c_n''$ with probability $\ge1-\delta'/2$, where $c_n''=O(\sqrt{\log(2/\delta')/n})$.

Step 3: Combining. By a union bound over the two concentration events (probability $\geq 1-\delta'$):

$$\begin{aligned}\widehat{\mathrm{osc}}_n - \widehat{\mathrm{osc}}_n^F &\geq \big(\mathrm{osc}_{\mathcal{K}_\alpha}(h_U) - c_n'\big) - \big(\mathrm{osc}_{\mathcal{K}_\alpha^F}(h_F) + c_n''\big) \\ &\geq \min_{j\neq 1} \Delta_j - (2r_1 + r_F) - (c_n' + c_n'') \\ &= \min_{j\neq 1} \Delta_j - 2r_1 - r_F - c_n,\end{aligned}$$

where $c_n = c_n' + c_n'' = O(\sqrt{\log(2/\delta')/n})$. □

A.7 Proof of Theorem 4.5 (certified gap improvement)

Proof. Write

$$\begin{aligned}C_U - C_F &= \big(\widehat{\mathrm{osc}}_n + 2M_U\varepsilon_U^*\big) - \big(\widehat{\mathrm{osc}}_n^F + 2M_F\varepsilon_F^*\big) \\ &= \big(\widehat{\mathrm{osc}}_n - \widehat{\mathrm{osc}}_n^F\big) + 2\big(M_U\varepsilon_U^* - M_F\varepsilon_F^*\big).\end{aligned}$$

By Lemma 4.4 (with probability $\geq 1-\delta'$), $\widehat{\mathrm{osc}}_n - \widehat{\mathrm{osc}}_n^F \geq \Delta := \min_j \Delta_j - 2r_1 - r_F - c_n$.
Case 1: $M_F\varepsilon_F^* \leq M_U\varepsilon_U^*$. Then $C_U - C_F \geq \Delta + 0 = \Delta > 0$ (by assumption $\Delta > 0$).
Case 2: $M_F\varepsilon_F^* > M_U\varepsilon_U^*$. The sufficient condition (17) gives $\Delta > 2(M_F\varepsilon_F^* - M_U\varepsilon_U^*)$, so

$$C_U - C_F \geq \Delta - 2(M_F\varepsilon_F^* - M_U\varepsilon_U^*) > 0.$$

In both cases, $C_F < C_U$. The function $C \mapsto 2/(1+e^C)$ is strictly decreasing, so

$$\underline{\gamma}_F = \frac{2}{1+e^{C_F}} > \frac{2}{1+e^{C_U}} = \underline{\gamma}_U.$$

The overall confidence is $\geq 1 - \delta_U - \delta_F - \delta'$ by a union bound over the three events: folded certificate (prob $\geq 1-\delta_F$), unfolded certificate ($\geq 1-\delta_U$), oscillation-gap concentration ($\geq 1 - \delta'$). □

B Boundary analysis

B.1 Fold boundary density for Gaussian mixtures

For the symmetric two-component Gaussian mixture $\pi = \frac{1}{2}\mathcal{N}(\mu_1, \sigma^2 I_d) + \frac{1}{2}\mathcal{N}(\mu_2, \sigma^2 I_d)$ with $\mu_1 = (\delta/2, 0, \dots, 0)$, $\mu_2 = -\mu_1$, and reflection fold boundary $\partial D = \{\theta_1 = 0\}$:

The density on the fold boundary is

$$\pi(0, \theta_{2:d}) = \frac{1}{2}(2\pi\sigma^2)^{-d/2} \exp\Big(-\frac{\delta^2/4 + \|\theta_{2:d}\|^2}{2\sigma^2}\Big) \cdot 2 = (2\pi\sigma^2)^{-d/2} \exp\Big(-\frac{\delta^2/4 + \|\theta_{2:d}\|^2}{2\sigma^2}\Big).$$

The peak density (at μ_1) is

$$\pi(\mu_1) = \frac{1}{2}(2\pi\sigma^2)^{-d/2} + \text{negligible}.$$

The ratio at the midpoint $\theta = 0$ is

$$\frac{\pi(0)}{\pi(\mu_1)} \approx 2\exp\Big(-\frac{\delta^2}{8\sigma^2}\Big).$$

For $\delta = 6\sigma$: ratio $\approx 2\exp(-4.5) \approx 0.022$.

This confirms that the fold boundary lies in a low-density valley, with density approximately 2% of the peak.

B.2 Condition for $\mathcal{K}_\alpha^F \cap \partial D = \varnothing$

The HPD set $\mathcal{K}_\alpha^F = \mathcal{K}_\alpha \cap D$ does not touch the fold boundary if and only if $\mathcal{K}_\alpha \cap \partial D = \varnothing$. For the Gaussian mixture above, $\mathcal{K}_\alpha$ consists of two balls of radius $R_\alpha = \sigma\sqrt{\chi^2_{d,1-\alpha}}$ centred at $\pm\mu_1 = (\pm\delta/2, 0, \dots, 0)$. The closest point of $\mathcal{K}_\alpha^{(1)}$ to ∂D has $\theta_1 = \delta/2 - R_\alpha$. The separation condition is

$$\mathcal{K}_\alpha^F \cap \partial D = \varnothing \quad\Longleftrightarrow\quad \delta/2 > R_\alpha \quad\Longleftrightarrow\quad \delta > 2\sigma\sqrt{\chi^2_{d,1-\alpha}}.$$

d	$\chi^2_{d,0.95}$	$2\sigma\sqrt{\chi^2}$	$\delta = 6\sigma$ OK?
2	5.99	4.90	Yes
5	11.07	6.65	No
10	18.31	8.56	No
20	31.41	11.21	No

For $\delta = 6\sigma$, the condition holds only at $d = 2$. At $d \geq 5$, the HPD set touches the fold boundary. However, the experiments in Section 5.3 show strong certified improvement at all dimensions, indicating that the boundary-touching case is handled adequately by the quotient-metric covering (Section 3.4).

B.3 Stabiliser analysis for permutation folding

For label-switching with $G = S_m$, the fundamental domain is the sorted chamber $D = \{(\theta^{(1)}, \dots, \theta^{(m)}) : \theta_1^{(1)} \leq \cdots \leq \theta_1^{(m)}\}$.

A point $z \in D$ has non-trivial stabiliser if and only if two or more blocks share the same value of the sort coordinate: $\mathrm{Stab}(z) \supsetneq \{e\} \iff \exists\, j \neq k : z_1^{(j)} = z_1^{(k)}$.

For well-separated components (centres $c_1 < c_2 < \cdots < c_m$ with $c_{j+1} - c_j \gg \sigma$), the HPD set $\mathcal{K}_\alpha^F$ is concentrated near the sorted centre $(c_1, \dots, c_m)$, which has trivial stabiliser. The fixed-point stratum $\{z_1^{(j)} = z_1^{(k)}\}$ lies at distance $\geq (c_{j+1} - c_j)/2 - R_\alpha$ from $\mathcal{K}_\alpha^F$, which is positive for sufficiently separated components.

In this case Assumption 3.7 holds and $h_{\max} = 1$. When components are not well separated (as in the structural identification example of Section 6), $h_{\max}$ may exceed 1 and the covering condition uses the corrected mass factor $(s/h_{\max})\, \pi^D_{\min}\, V_d\, \varepsilon^d$.

C Experimental details

C.1 Architecture and hyperparameters

All flows use spectrally normalised RealNVP with tanh activations and scale clip $c = 0.7$. The architecture scales with dimension:

d	Layers L	Hidden dim h	Epochs	Batch size
≤ 4	8	64	2000	512
5–6	10	128	2000	512
8–12	12	128	3500	512
20	16	256	3500	512

Training uses Adam with learning rate 10^{-3} and annealed oscillation/gradient regularisation (FolT-OG-Anneal, as in Hu 2026a). The regularisation weight is annealed from 0 to its final value over the first 30% of training.

The folded and unfolded pipelines use identical architecture and hyperparameters for each d. Training samples are drawn from the target's exact sampler (for synthetic targets) or from a long-run random-walk Metropolis chain (for the structural identification application).

C.2 Certification protocol

Certification uses $n = 20\,000$ independent samples from the target (folded samples projected onto D for the folded pipeline). Quantile-core certificates are computed at $\rho \in \{0.025, 0.05, 0.10, 0.25\}$ with DKW-corrected confidence $\delta = 0.05$. The primary metric reported in all tables is the $\rho = 0.05$ certificate.

The gradient supremum M_F is estimated as the empirical maximum of $\|\nabla h_F(Z_i)\|$ over the certification samples, computed via automatic differentiation.

C.3 MCMC protocol

Independence Metropolis–Hastings with 4 chains of 5 000 steps each (10 000 for the RPS baseline comparison), burn-in equal to 20% of the chain length. Effective sample size (ESS) is estimated via batch means with batch size $\sqrt{n_{\text{chain}}}$.

For the structural identification application (Section 6), training samples are generated by a preliminary random-walk Metropolis run of 200 000 steps with adaptive step size, thinned to 50 000 samples.

C.4 Full quantile-core tables

Gaussian mixture, dimension scaling.

	QC $\underline{\gamma}$ at $\rho =$				
d	0.025	0.05	0.10	0.25	Full γ
Unfolded					
2	0.216	0.402	0.608	0.810	$\sim 10^{-3}$
5	0.031	0.094	0.260	0.565	$\sim 10^{-5}$
10	0.005	0.016	0.065	0.290	$\sim 10^{-7}$
20	0.005	0.016	0.062	0.280	$\sim 10^{-5}$
FolT-MCMC					
2	0.890	0.936	0.968	0.990	0.06–0.18
5	0.895	0.941	0.970	0.991	0.06–0.18
10	0.870	0.922	0.958	0.986	0.06–0.18
20	0.850	0.902	0.950	0.983	0.06–0.18

Label switching.

		QC $\underline{\gamma}$ at $\rho =$			
Config	s	0.025	0.05	0.10	0.25
Unfolded					
m=2, p=2	2	0.142	0.366	0.571	0.790
m=3, p=2	6	0.019	0.066	0.185	0.486
m=3, p=4	6	0.010	0.036	0.111	0.380
m=4, p=2	24	0.001	0.004	0.018	0.115
FolT-MCMC					
m=2, p=2	2	0.620	0.820	0.905	0.965
m=3, p=2	6	0.450	0.685	0.830	0.935
m=3, p=4	6	0.445	0.683	0.828	0.930
m=4, p=2	24	0.380	0.624	0.785	0.910

Banana diagnostic ($d = 2$, $s = 2$).

	QC $\underline{\gamma}$ at $\rho =$			
	0.025	0.05	0.10	0.25
Unfolded	0.064	0.272	0.537	0.797
FolT-MCMC	0.135	0.320	0.523	0.759

Note that at $\rho = 0.10$ and 0.25, the unfolded certificate slightly exceeds the folded one for the banana target. This reflects the fold-boundary residual spike entering the ρ-core at larger ρ, consistent with the diagnostic analysis in Section 5.2.

C.5 Condition verification for Theorem 4.5

We verify conditions (M′), (W), (R), and (S) empirically for the Gaussian mixture ($d = 10$, $s = 2$) and label-switching ($m = 4$, $p = 2$, $s = 24$) experiments.

Quantity	Mixture d=10	Label m=4
$\min_j \Delta_j$ (from QC osc gap)	≥ 4.69	≥ 5.35
$2r_1$ (folded within-mode osc)	≤ 0.32	≤ 1.58
r_F (quotient proposal residual)	≈ 0	≈ 0
$\Delta = \min_j \Delta_j - 2r_1 - r_F$	≥ 4.37	≥ 3.77
$2(M_F \varepsilon_F^* - M_U \varepsilon_U^*)^+$	≈ 0	≈ 0
Condition (S) satisfied?	Yes	Yes
Δ / C_U	0.90	0.87

In both cases, the cross-mode deficiency Δ_j is much larger than the folded within-mode oscillation r_1, and the quotient proposal residual r_F is negligible. The sufficient condition (S) is easily satisfied, confirming that the certified improvement in Theorem 4.5 holds.